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ORIGINAL ARTICLE

Tumor cells express pauci- and oligomannosidic N-glycans in glycoproteins recognized by the mannose receptor (CD206)

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Abstract

The macrophage mannose receptor (CD206, MR) is an endocytic lectin receptor which plays an important role in homeostasis and innate immunity, however, the endogenous glycan and glycoprotein ligands recognized by its C-type lectin domains (CTLD) have not been well studied. Here we used the murine MR CTLD4–7 coupled to the Fc-portion of human IgG (MR-Fc) to investigate the MR glycan and glycoprotein recognition. We probed 16 different cancer and control tissues using the MR-Fc, and observed cell- and tissue-specific binding with varying intensity. All cancer tissues and several control tissues exhibited MR-Fc ligands, intracellular and/or surface-located. We further confirmed the presence of ligands on the surface of cancer cells by flow cytometry. To characterize the fine specificity of the MR for glycans, we screened a panel of glycan microarrays. Remarkably, the results indicate that the CTLD4-7 of the MR is highly selective for specific types of pauci- and oligomannose N-glycans among hundreds of glycans tested. As lung cancer tissue and the lung cancer cell line A549 showed intense MR-Fc binding, we further investigated the MR glycoprotein ligands in those cells by immunoprecipitation and glycoproteomic analysis. All enriched glycoproteins, of which 42 were identified, contained pauci- or oligomannose N-glycans, confirming the microarray results. Our study demonstrates that the MR CTLD4-7 is highly selective for pauci- and oligomannosidic N-glycans, structures that are often elevated in tumor cells, and suggest a potential role for the MR in tumor biology.

Keywords Mannose receptor · Glycans · Cancer tissue · Ligands · Glycan microarray · Paucimannose · Oligomannose · Glycoproteomics · C-type lectins · Glycan-binding proteins

Abbreviations

AEAB	2-Amino-N-(2-aminoethyl)-benzamide
ACN	Acetonitrile
CFG	Consortium for functional glycomics
CR domain	Cysteine-rich domain
CTLD	C-type lectin domain
FA	Formic acid
Fab	Fragment antigen binding
Fuc	Fucose
Gal	Galactose
GalNAc	N-acetylgalactosamine
Glc	Glucose
GlcNAc	N-acetylglucosamine
Hex	Hexose
HexNAc	N-acetylhexosamine
HI FBS	Heat-inactivated fetal bovine serum
IgG	Immunoglobulin G
LC–MS	Liquid chromatography-mass spectrometry
mAb100	Monoclonal antibody 100-4G11-A
Man	Mannose
MR-Fc	Mannose receptor-fragment crystallizable
MR, CD206	Mannose receptor
NeuAc	N-acetylneuraminic acid
NSCLC	Non-small cell lung cancer
PaTu-T	Pa-Tu-8988 T cells
PSG	Penicillin/Streptomycin/Glutamine
TAM	Tumor-associated macrophages

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Introduction

The macrophage mannose receptor (CD206, MR) is an endocytic lectin receptor expressed primarily in macrophages, dendritic cells and some endothelial cells. It plays an important role in endocytosis, antigen presentation and induction of immune responses [1]. The MR consists of three domains—the N-terminal cysteine-rich (CR) domain, followed by a fibronectin type-II domain, and eight contiguous C-type lectin domains (CTLD). Both the CR and CTLD are involved in glycan-binding. The ligands of the CR domain have been identified as sulfated galactose (Gal) or N-acetylgalactosamine (GalNAc) in which sulfate occurs at position C3 or C4 [2–4]. The glycan-binding specificity of the CTLD, however, is less well understood, although it is reported to bind mannose (Man), fucose (Fuc), N-acetylglucosamine (GlcNAc) and glucose (Glc)[5–8]. Out of the eight CTLD repeats of the MR only CTLD4 has the ability to bind glycans when expressed as an individual domain. When expressed with CTLD5, they account for all CTLD glycan-binding affinity [5, 6]. Several pathogen binding studies illustrate the capacity of the MR-CTLD4–7 to bind mannose-containing glycans [9, 10]. However, the binding-profile of the MR in terms of specificity or preference for its glycan ligands is incompletely understood, especially with regard to endogenous glycan ligands potentially expressed in normal and cancerous human cells.

Proteins carrying oligomannose and paucimannose N-glycans, which can serve as ligands for the MR, have been identified to be elevated in a variety of cancers and to play key roles in inflammation and cancer [reviewed in [11, 12]]. Several lectin receptors present on innate immune cells have the ability to promote immune suppression in the tumor microenvironment [13]. Among these lectins, the MR also has the capacity to induce anti-inflammatory responses [10, 14].

The MR is a prominent lectin present on macrophages, which play critical roles in the regulation of anti-tumor immune responses [15, 16]. Macrophages include a diversity of cell-subsets that range from inflammatory (called classically activated or M1) macrophages to anti-inflammatory (called alternatively activated or M2) macrophages [17]. They abundantly infiltrate into tumors (tumor-associated macrophages—TAMs), and dependent on the tumor environment they can polarize towards a subset of cells with M2-like properties, thereby impairing antitumor immunity which is associated with a poor prognosis in different animal models and human patients [18–20]. Since M2 macrophages commonly express the MR [21, 22], studying the interaction of this lectin receptor with tumor cells, and its putative role in allowing tumor progression by immune suppression, is important.

To date, little is known about the glycan- and glycoprotein-binding specificity of the MR to endogenous ligands. However, Stahl and colleagues who initially discovered the MR in the 1970s demonstrated its binding to endogenous lysosomal enzymes, such as β-glucuronidase and N-acetyl-β-D-glucosaminidase in alveolar macrophages [7] and Kupffer cells [23]. Additional identified MR ligands include neutrophil-derived myeloperoxidase [24], tissue plasminogen activator [25], type I procollagen [26], murine thyroglobulin [27] and other lysosomal enzymes [27, 28]. The MR binding is suggested to be mediated via the CTLD since it can be reduced or diminished in the presence of mannose-containing glycans or glycoconjugates.

Here, we report the presence, and cell- and tissue-specific localization of MR-ligands in human cancer tissue microarrays containing a variety of different types of tumors and control tissues using a recombinant form of the MR (MR-Fc). MR-Fc is a hybrid protein containing CTLD4-7 from the murine MR fused to the human immunoglobulin G (IgG) Fc fragment. We confirmed the surface expression of MR glycan ligands on different tumor cell types and analyzed the endogenous glycoproteins interacting with MR using glycoproteomics. In addition, we characterized the specific glycan-binding profile of the MR using a wide variety of glycan microarrays containing endogenous and pathogen-derived glycan ligands. Collectively our studies demonstrate a highly preferred binding of the MR to pauci- and oligomannose glycans. The identification of MR glycan ligands in cancer cells will facilitate further studies to understand the role of this receptor in cancer cell recognition and putative immune evasion and might reveal a new set of potential therapeutic targets.

Methods and materials

Materials

Ammonium bicarbonate, $CaCl_2$, EDTA, ethanol, H_2O_2, LC–MS grade acetonitrile (ACN), LC–MS grade formic acid (FA), $MgCl_2$ were purchased from Sigma Aldrich, Millipore. Xylene and hematoxylin were obtained from Electron Microscopy Sciences and BSA (Fraction V), Glycine, NaCl, sodium citrate, Triton X-100, Tween 20 from Fisher Bioreagents. Protease inhibitor was purchased from Roche, Tris–HCl from Promega and citric acid from Mallinckrodt.

Cells and culture conditions

The human pancreatic adenocarcinoma cancer cell line Pa-Tu-8988 T (PaTu-T) was purchased at DSMZ culture bank (Braunschweig, Germany). The human colorectal cancer cell lines HCT116 and SW620 were obtained from the

Department of Pathology, VU Medical Center Amsterdam. The human non-small cell lung cancer cell line A549, the human epidermoid carcinoma cells A431, the breast cell cancer cell line SKBR3 and the melanoma cell line MelJKO were obtained from the Department of Medical Oncology (VU Medical Center Amsterdam, Netherlands).

PaTu-T cells were cultured in RPMI-1640 (Gibco), supplemented with 10% heat-inactivated fetal bovine serum (HI FBS) and 1% Penicillin/Streptomycin/Glutamine (PSG) at 37 °C, in a 5% CO_2 humidified atmosphere. A431, A549, SW620, SK-BR3 and HCT116 cells were cultured in DMEM high glucose (Gibco) with 10% HI FBS and 1% PSG, and MelJKO cells in IMDM supplemented with 10% HI FBS and 1% PSG. To prepare recombinant MR, Human embryonic kidney cells (HEK)293 T cells transfected with MR-Fc and were cultured in RPMI with 10% HI FBS, 1% PS, 10 mM HEPES containing non-essential amino acids. A549 cells for subsequent MR-Fc glycoprotein ligand pull down were harvested with 1 mM EDTA in PBS.

MR-Fc expression and purification

HEK293T cells were transfected with MR-Fc DNA consisting of the murine MR CTLD4-7 fused to a human IgG Fc-portion (CRD4-7Fc; kind gift from L. Martinez Pomares [27]). Transfection was performed with Lipofectamine LTX with Plus Reagent (Thermo Fisher Scientific) conform with the manufacturer´s guidelines. The cells were incubated with Lipofectamine complex at 37 °C in a CO_2 incubator for 24 h, and the medium refreshed after 24 h. After 9 days the supernatant containing the produced MR-Fc was collected and either used directly or after purification. MR-Fc was purified from cell culture supernatant using Protein A-agarose beads (Roche). After incubation with the beads at 4 °C on a rotator overnight, the beads were washed three times with PBS and MR-Fc was eluted using 0.1 M Glycine–HCl, pH 2.7, followed by immediate neutralization with 1 M Tris–HCl, pH 8.7. MR-Fc was stored at −20 °C until further use.

Glycan microarray analysis

Glycan microarrays were screened using a standard protocol as published elsewhere [29]. Briefly, MR-Fc binding was performed in TSM binding buffer [TSM (20 mM Tris–HCl, pH 7.4, 150 mM NaCl, 2 mM $CaCl_2$, 2 mM $MgCl_2$) + 0.05% Tween 20 + 1% BSA] for 1 h at RT on shaker, followed by four washes with TSM + 0.05% Tween 20 and incubation with a secondary antibody (goat anti-human IgG Alexa-Fluor 488-conjugated, 5 µg/ml, Jackson ImmunoResearch) in TSM binding buffer for 1 h at RT on shaker. The slides were washed four times with TSM + 0.05% Tween 20, followed by TSM and then H_2O. As a control, the binding was performed in the presence of 5 mM EDTA instead of $CaCl_2$. The arrays were scanned using a GenePix 4300A scanner (Molecular Devices) at 488 nm and data readout was performed using the GenePix Pro software package.

The following glycan microarrays were screened for binding under the same conditions as mentioned above: MR-Fc binding was analyzed on the Consortium for Functional Glycomics (CFG) glycan array [29] (version 5.4) at 16 µg/mL; recently generated oligomannose array [30] at 30 µg/mL; mannose-6-phosphate glycan microarray [31] at 4 µg/mL; microbial glycan microarray (MGM) [32] at 0.8 µg/mL; and the *Schistosoma mansoni* glycan and shotgun microarrays [33, 34] at 4 µg/mL. The full-length human recombinant MR (His-tag, R&D Systems) was analyzed for binding on the oligomannose array at 10 µg/mL with anti-penta-His Alexa-Fluor 488 conjugate (Qiagen) at 5 µg/mL, and mAb100-4G11-A (here abbreviated mAb100) [35] at 1:200 dilution with goat anti-mouse IgM secondary antibody at 5 µg/mL (Alexa-Fluor 488-conjugated, Invitrogen). Details about the preparation of these arrays can be found in the corresponding references.

Immunohistochemistry of tissue microarrays

Immunohistochemistry was performed on a formalin-fixed, paraffin-embedded tissue microarray containing 96 tissue cores (#BCN963B, US Biomax Inc.) adapted from a previously published protocol [36]. Slides were prepared for antigen retrieval by incubation at 60 °C for 30 min, followed by de-paraffinization in xylene for 2 × 30 s, rehydration in ethanol for 2 × 30 s, 95% ethanol for 30 s, 70% ethanol for 30 s and rinsing in water for 30 s. Antigen retrieval was achieved by heating the slides in 10 mM sodium citrate buffer pH 6.0 in a pressure cooker for 3 min after it reached full pressure. The slides were rinsed with water, treated with 3% H_2O_2 for 10 min and rinsed again with TBST [TBS (20 mM Tris, 300 mM NaCl, pH 7.2) + 0.1% Tween 20] for 3 × 5 min. Slides were blocked with 10% goat serum (Thermo Fisher Scientific) in TBST for 1 h at RT, followed by incubation with 20 µg/mL MR-Fc in 5% goat serum in TBST containing 5 mM $CaCl_2$ or 200 × dilution of mAb100 in 5% goat serum overnight at 4 °C. To determine non-specific binding another set of slides was incubated with 20 µg/mL MR-Fc in 5% goat serum in TBST containing 10 mM EDTA or 5 µg/mL mouse IgM (Bethyl Laboratories), respectively. For MR-Fc binding the following steps were performed with TBST containing 5 mM $CaCl_2$ and for all other samples only TBST was used. The next day the slides were washed 3 × in TBST for 10 min and incubated with the secondary antibody [for MR-Fc: HRP-conjugated goat anti-human IgG (H + L) (5 µg/mL, Jackson Immunology); for mAb100: HRP-conjugated goat anti-mouse IgM (5 µg/mL, Jackson Immunology)]. The slides were washed with TBST and developed with AEC single solution (Life Technology, Thermo Fisher Scientific).

Hematoxylin staining was performed for 4 min, followed by washing in TBST for 10 min and mounting of the slides using Vectashield (Vector Lab). Tissue slide pictures were taken using a VS120 Slide scanner (Olympus Lifesciences) at 20 × magnification and pictures were processed (background subtraction, contrast correction) using ImageJ [37].

The extent and intensity of the staining in the cell population of interest (tumor cells or normal tissue) were assessed visually by a certified pathologist (Dr. Jonathan N Glickman, co-author herein). H-scoring was applied for quantitation which is calculated as follows: H-score = (1 × percentage of cells with of weak staining, discernable only at high 400 × magnification) + (2 × percentage of cells with moderate staining, discernable at intermediate magnification) + (3 × percentage of cells with strong staining, visible at low 40 × magnification) [38]. The subcellular localization (membrane vs cytoplasmic) of staining was also noted.

Flow cytometry

Cells were harvested with trypsin–EDTA in their exponential growth phase. For binding studies, the different cell types were incubated with human MR-Fc with and without EGTA (10 mM), with anti-Man_3 antibody (mAb100)[35], human $IgG1_k$ (AdB Serotec, BioRad) and as a control fixable viability dye eFluor 780 (eBioscience) in TSM, to exclude dead cells, with 0.5% BSA for 1 h at 4 °C. Subsequently, the cells were incubated for 30 min with secondary antibody (goat anti-human IgG Fc FITC; Jackson ImmunoResearch Laboratories) for MR-Fc and goat anti-mouse IgM (μ chain; Alexa-Fluor 647; Invitrogen) for mAb100. The cells were analyzed on a Dako Cytomation Cyan ADP flow cytometer (Beckman Coulter) or FACS-Calibur flow cytometer (Becton–Dickinson). Data were analyzed with FlowJo software version 10 for Windows (Tree Star).

Data are presented as the mean ± SEM. Performed data analysis with SPSS Statistics version 22 (IBM). Data were analyzed using One-Way ANOVA with Dunnett´s or Tukey´s multiple comparison test. Significant values are indicated with * for $p \leq 0.05$ and ** for $p \leq 0.01$ and *** for $p \leq 0.001$.

MR-Fc glycoprotein enrichment and in-gel proteolysis

Approximately 5×10^7 A549 cells were lysed in lysis buffer (TBS containing 0.1% triton X-100, protease inhibitor) for 30 min on ice, followed by sonication using a sonicator probe (Qsonica) and 3 cycles of 20 Amp for 10 s on ice. Cell lysate was centrifuged at 10,000 × *g* for 20 min at 4 °C and supernatants were collected. One mg of cell lysate was incubated with 15 μg MR-Fc in the presence of 10 mM $CaCl_2$ and as a control with 100 mM EDTA at 4 °C on a shaker overnight. The next day 50 μL protein G magnetic beads (Dynabeads, Thermo Fisher) were added and incubated for another 2 h at 4 °C on a shaker. MR-Fc ligands were enriched on a magnetic holder, by washing the beads three times with TBS containing 10 mM $CaCl_2$ and eluting with TBS containing 100 mM EDTA. The eluents of two samples were combined and the experiment was performed in triplicates. After adding 4 × reducing Laemmli buffer the samples were subjected to SDS-PAGE to run into the gel for ~ 2 cm. Each lane was cut into 5 bands and subjected to in-gel digestion as described before [39]. Protein bands were digested with 40 μL of 0.0025 μg/μL trypsin (sequencing grade, Promega) in 25 mM ammonium bicarbonate overnight at 37 °C. The next day the supernatant was collected and 50 μL of 50% acetonitrile were added and incubated for 10 min on a shaker. Both supernatants were combined and dried down in a speed vac concentrator. The samples were stored at − 20 °C until further use.

LC–MS glycopeptide analysis and data analysis

Liquid chromatography-mass spectrometry (LC–MS) was performed on an Ultimate 3000 nano-LC coupled to an Orbitrap Fusion Lumos mass spectrometer (both Thermo Fisher) as described elsewhere [40]. Samples were dissolved in 14 μL of 0.1% formic acid (FA) in H_2O and 4 μL were loaded onto a C18 precolumn (C18 PepMap 100, 300 μm × 5 mm, 5 μm, 100 Å, Thermo Scientific) with 15 μL/min solvent A (0.1% FA in H_2O) for 3 min and separated on a C18 analytical column (PicoFrit 75 μm ID × 150 mm, 3 μm, New Objective) using a linear gradient of 2% to 45% solvent B (80% acetonitrile, 0.1% FA) over 60 min, followed by 45% to 90% B over 5 min at 400 nL/min. The mass spectrometer was operated under the following conditions: The ion source parameters were 2000 V spray voltage and 200 °C ion transfer tube temperature. MS scans were performed in the orbitrap at a resolution of 60,000 within a scan range of *m/z* 400 – *m/z* 1600, a RF lens of 30%, AGC target of 1e5 for a maximum injection time of 50 ms. The top 15 precursors were selected for MS^2 in a data-dependent manner, within a mass range of *m/z* 550 – *m/z* 1600 and a minimum intensity threshold of 1e5 and an isolation width of 2 m/z. HCD is performed in stepped collision energy mode of 30% (± 5%) and detected in the orbitrap with a resolution of 30,000 with the first mass at *m/z* 120 an AGC target of 2e5 and a maximum injection of 250 ms.

Data analysis was performed against the human Swiss prot database (v 2016–05-11) using SEQUEST through proteome discoverer (version 2.1.0.81, Thermo Fisher Scientific) under the following settings: Trypsin (2 missed cleavage sites), precursor mass tolerance of 10 ppm and fragment mass tolerance of 0.02 Da. Dynamic modifications were oxidation on Met, deamidation on Asn, phosphorylation on Ser/Thr and N-terminal acetylation. Static modifications were

carbamidomethyl on Cys. All identified proteins were used as a database for subsequent glycopeptide data analysis.

Glycopeptide identification was performed using Byonic version 3.5 (Protein Metrics Inc.). Trypsin was set as protease with a maximum of two missed cleavage sites, the precursor and fragment mass tolerance was set to 10 ppm. The glycan database was "N-glycan 309 mammalian" and "O-glycan 9 common". The following modifications were allowed: carbamidomethyl (Cys; fixed), oxidation (Met; variable common 1), pyroglutamine on N-term (Gln, variable rare 1), ammonia-loss N-term (Cys; variable rare 1), acetylation N-term (variable rare 1), deamidation (Asn, variable common 1), formylation N-term (variable rare 1). Glycopeptides with a score above 250 were selected and further manually inspected.

Glycoproteins were identified by at least two unique peptides and a glycopeptide in two out of three replicates, enriched in a calcium-dependent manner. Only the glycoprotein Cathepsin D was also identified on a peptide level, but not glycopeptide level, in the EDTA control. Since the number of peptide spectra matches was only minor compared to the calcium-dependent enrichment it was still included in the list of identified glycoproteins.

Results

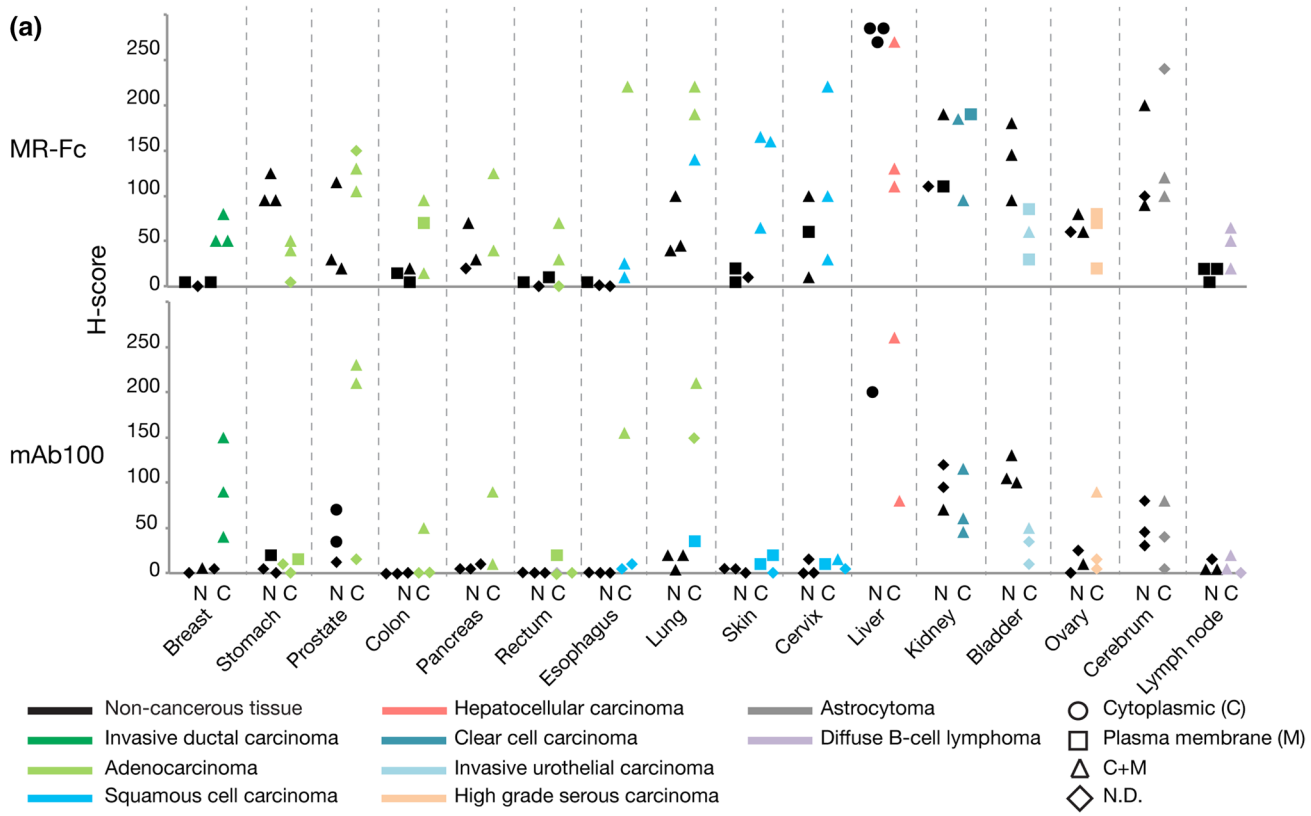
Cell- and tissue-specific binding of MR-Fc to cancer and control tissue

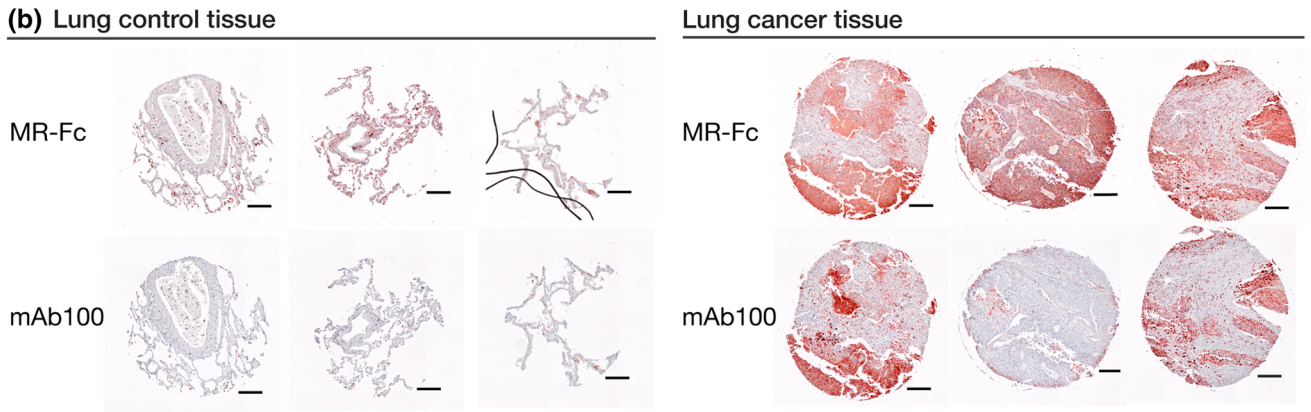
Recent findings demonstrated the presence of pauci- and oligomannosidic N-glycans in several cancer types, which prompted our studies on whether the MR can bind to such cells [11, 12]. To explore the presence of MR ligands across different control and cancer tissues, we screened a tissue microarray with a hybrid MR protein containing the mannose-binding domains CTLD4-7 fused to an IgG Fc portion (MR-Fc) as an IHC probe. In addition, we used a monoclonal antibody mAb100, which binds $Man_3GlcNAc_2$ [35] to identify paucimannosidic N-glycans within the tissues. The tissue microarray comprises 16 different tissue types with each three cancer and control tissue cores from separate individuals. The binding was quantified by H-scoring (Fig. 1a). MR-Fc and mAb100 secondary controls showed no non-specific binding except for mAb100 in normal liver tissue, which was excluded from the dataset (Suppl. Table S1). When present, MR-Fc binding was observed either at the plasma membrane or to both the plasma membrane and cytoplasm, except for hepatocytes in control liver tissue, which showed only cytoplasmic binding. MR-Fc bound to 14 out of the 16 control tissue types in a tissue and cell type-specific manner to varying degrees (Fig. 1, Suppl. Figure S1-S4, Suppl. Table S1). Epithelial cells were observed to express MR-Fc ligands in several control tissues, including the colon, prostate, bladder (urothelium) and kidney tubules. In addition, MR-Fc ligands were found in bladder, cervix and ovarian stroma, skin basal keratinocytes, neurons and glial cells in the cerebrum, glandular cells in the stomach and type II pneumocytes in the lung, muscle cells in the esophagus, as well as acinar and islet pancreatic cells (Fig. 1, Suppl. Table S1). Inflammatory cells in multiple organs and in lymph nodes also express MR-Fc ligands. The monoclonal antibody mAb100 showed a similar tissue distribution of staining, in a subset of the MR-Fc stained cells, with particularly strong staining in liver, kidney and bladder control tissue (Fig. 1a, Suppl. Table S1). Collectively, the results of MR-Fc and mAb100 binding indicate the presence of paucimannose N-glycans in those tissues.

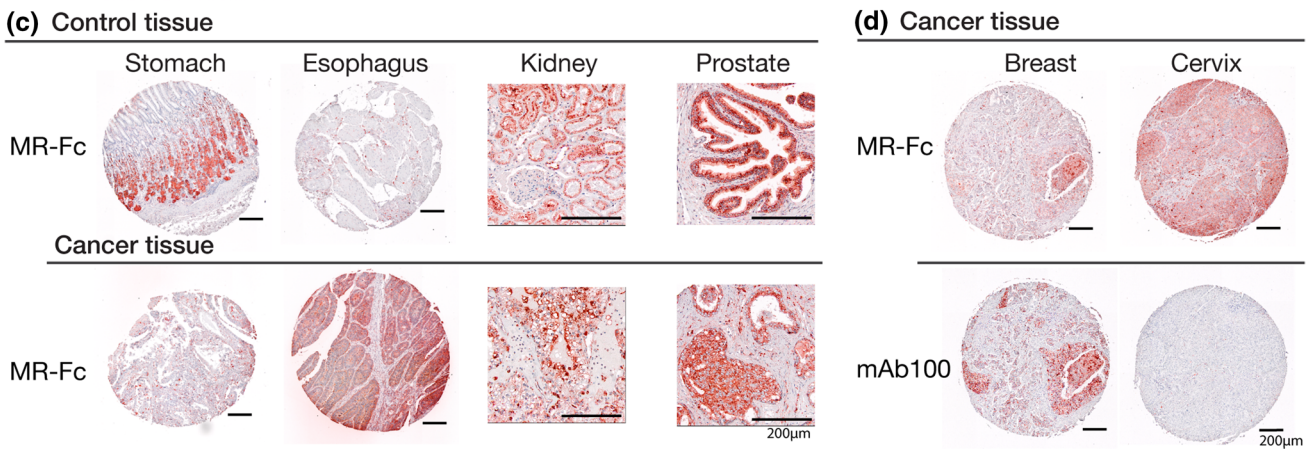
Interestingly, all 16 cancer tissue types showed varying degrees of MR-Fc binding, and in some tumor types with an immense variation among the three individual tissue samples (Fig. 1, Suppl. Figure S1–S4, Suppl. Table S1). The most striking difference of MR-Fc binding between cancer and control tissue was found in the lung (Fig. 1b). While the control tissue only showed modest staining in a few type II pneumocytes and inflammatory cells, the lung adenocarcinoma and squamous cell carcinoma were strongly stained with both MR-Fc and mAb100. In addition, breast invasive ductal carcinoma, prostate adenocarcinoma and esophagus adenocarcinoma showed more intense staining with MR-Fc and mAb100 compared to control tissues (Fig. 1a, c, d, Suppl. Table S1). A similar observation was made for MR-Fc binding to skin squamous cells carcinoma compared to control tissue (Fig. 1a, Suppl. Table S1), however, little mAb100 binding was observed. The opposite result was observed in the stomach tissue, with strong binding of MR-Fc to normal oxyntic glandular cells and little staining of adenocarcinoma (Fig. 1a, c). Similarly, bladder urothelium and stroma control tissue showed higher MR-Fc and mAb100 binding compared to invasive urothelial carcinoma (Fig. 1a, Suppl. Table S1). Two of three hepatocellular carcinoma samples showed less intense staining with MR-Fc compared to control liver, which was intensely stained (Fig. 1a, Suppl. Table S1). Similar MR-Fc and mAb100 binding in cancer and control were observed for kidney, cerebrum, cervix and ovary tissue (Fig. 1, Suppl. Table S1).

MR-Fc and mAb100 ligands on the surface of cancer cell lines

The tissue microarray studies show the presence of MR-Fc ligands in several cancer tissues, with indications of their expression on cell membranes. To further explore the potential surface localization of MR ligands on cancer cells, we screened a selection of cancer cell lines for MR-Fc and mAb100 binding using flow cytometry. Binding studies


(a)
MR-Fc
mAb100
H-score
250
200
150
100
50
0
N C
Breast
Stomach
Prostate
Colon
Pancreas
Rectum
Esophagus
Lung
Skin
Cervix
Liver
Kidney
Bladder
Ovary
Cerebrum
Lymph node
Non-cancerous tissue
Invasive ductal carcinoma
Adenocarcinoma
Squamous cell carcinoma
Hepatocellular carcinoma
Clear cell carcinoma
Invasive urothelial carcinoma
High grade serous carcinoma
Astrocytoma
Diffuse B-cell lymphoma
Cytoplasmic (C)
Plasma membrane (M)
C+M
N.D.


(b) Lung control tissue
Lung cancer tissue
MR-Fc
mAb100


(c) Control tissue
Stomach
Esophagus
Kidney
Prostate
MR-Fc
Cancer tissue
MR-Fc
200µm
(d) Cancer tissue
Breast
Cervix
MR-Fc
mAb100
200µm

◂**Fig. 1** Tissue microarray screening for ligands of MR-Fc and the Man3-specific antibody mAb100 using immunohistochemistry. A total of 96 tissue cores from 16 tissue types (each three cancer and control) were screened. **a** Quantitation of MR-Fc and mAb100 binding by H-scoring and cellular localization of the staining. **b** Lung control and cancer tissue staining with MR-Fc and mAb100. **c** MR-Fc binding of cancer and control tissue from stomach, esophagus, kidney and prostate. **d** MR-Fc and mAb100 binding in breast and cervix tissue. Overall, MR-Fc and mAb100 ligands are expressed in a cell- and tissue-specific manner in cancer and control. Scale bars are 200 µm

were performed at 4 °C to prevent internalization and ensure cell surface staining. All tested cancer cell lines, including adenocarcinoma from breast (SK-BR3), colon (SW620), melanoma (MelJKO), lung carcinoma (A549), colorectum (HCT116), pancreas (PatuT), and epidermoid tissue (A431) demonstrated significant MR-Fc binding to the cell surface, and with exception of MelJKO, the cells were also significantly bound by mAb100 (Fig. 2). Notably, among all cell lines tested, we observed that MR-Fc binding was the most intense in the lung carcinoma cell line A549. These results are in line with our observations from the tissue microarray and confirm that many cancer cells carry MR-Fc ligands on their cell surface.

MR-Fc binds pauci- and oligomannose glycans on glycan microarrays

To gain further information about the glycan-binding specificity and preference of the mannose-recognizing domains of the MR-Fc protein, we screened its binding to a variety of glycans on several glycan microarrays. On the CFG array [29] (version 5.4), which contains 585 structures representing mammalian glycans and glycan fragments, MR-Fc bound almost exclusively to mannose-terminating glycans, with a high preference for $Man_3GlcNAc_2$ and the Man3 trisaccharide of the N-glycan core (Fig. 3a, Suppl. Table S2). We also observed binding to oligomannose structures such as $Man_7GlcNAc_2$, but no binding was detected toward the various fucose- or GlcNAc-terminating glycans on the array. The analysis of MR-Fc binding to the mannose-6-phosphate glycan microarray [31] revealed that mannose binding is exclusive to non-phosphorylated oligomannose glycans (Fig. 3b, Suppl. Table S3). On this array MR-Fc shows binding to oligomannose glycans ranging from $Man_5GlcNAc_2$ to $Man_9GlcNAc_2$, including the $Man_7GlcNAc_2$.D1 isomer (#3 in Fig. 3b), however, no binding was observed to the $Man_7GlcNAc_2$.D2 isomer (#4 in Fig. 3b).

In order to obtain a more specific view of the MR-Fc mannose-binding pattern, we screened the MR-Fc on our oligomannose array Set 1 [30]. This array contains 81 pauci- and oligomannose N-glycans or portions of these glycans, representing a wide range of mannose-terminating glycans and their intermediates found in mammals (Suppl. Table S4). MR-Fc binds to glycans ranging from paucimannose glycans, such as the N-glycan core $Man_3GlcNAc_2$-related structures, to oligomannose-type glycans up to $Man_8GlcNAc_2$-related structures. Interestingly, N-glycan core structures with α1-3-linked Man on the 6-arm of the core (e.g. #48, #50 in Fig. 3c) seemed to be less preferred binders than α1-6-linked Man (e.g. #18, #23 in Fig. 3c). Considering the oligomannose intermediate structures the same pattern can be observed with preferred binding to glycans with α1-6-linked Man (+ optional α1-2-linked Man) to the 6-arm of the core (#9–23 and #35–43 in Fig. 3c) compared to the corresponding glycans with α1-3-linked Man (+ optional α1-2-linked Man) extensions to the 6-arm or α1-2-linked Man to the 3-arm of the core (#24–34, #44–59 in Fig. 3c). In accordance with the CFG array there was no binding to complex-type glycans with terminating GlcNAc, Gal or N-acetylneuraminic acid (NeuAc) (#82–84 in Fig. 3c). Overall our results from the glycan microarray analyses revealed binding of the MR-Fc predominantly to a subset of mannose-terminating glycans.

The full-length recombinant human mannose receptor was also applied to the oligomannose microarray, resulting in a binding pattern highly similar to (murine-derived) MR-Fc (Fig. 3d). Consequently, MR-Fc was found to be an excellent probe to study the glycan binding of the human MR CTLD4-7. For the studies on glycan interactions, we also examined mAb100, which we screened on the oligomannose array (Fig. 3e). Overall, the data show that mAb100 glycan ligands represent a subset of the MR-Fc glycan ligands. Notably, this distribution is in accordance with our findings using cancer tissue microarrays (Fig. 1).

Since the MR-CTLD highly prefers binding to paucimannosidic N-glycans, which can be found in some pathogens, we screened some of the pathogen-derived microarrays available, e.g. a *Schistosoma mansoni* glycan and shotgun array [33, 34] and a microbial glycan array [32]. Remarkably, the main MR glycan ligands found in the *S. mansoni* glycan array were pauci- and oligomannosidic glycans, with or without core-fucosylation and/or core-xylosylation. In addition, MR-Fc showed binding to the chitin-like oligosaccharide $GlcNAc_3$-AEAB on the *S. mansoni* glycan array (Suppl. Table S5). In the microbial glycan array, MR bound almost exclusively to yeast mannan and galactomannan (Suppl. Table S6), indicating the ability of the MR to bind to highly mannosylated structures, but also perhaps having a weak affinity for terminal GlcNAc within some glycans.

MR glycoprotein ligands in the lung cancer cell line A549 contain pauci- and oligomannose glycans

With consideration for the small set of tissue samples, both lung cancer primary tissue (Fig. 1a, b) and the lung cancer cell line A549 (Fig. 2) were characterized by enhanced

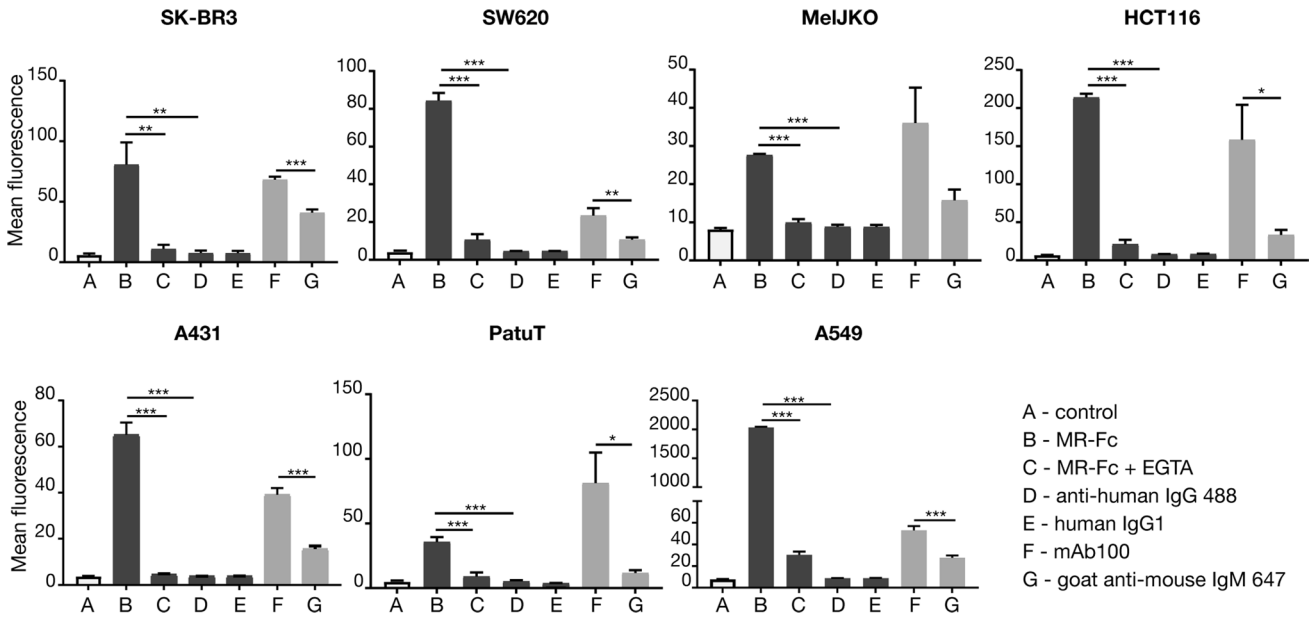


Fig. 2 Cell surface binding of MR-Fc (bar B) and mAb100 (bar F) to cancer cell lines analyzed by flow cytometry. Analyzed cell lines were the adenocarcinoma from breast (SK-BR3), colon (SW620), melanoma (MelJKO), colorectum (HCT116), epidermoid tissue (A431), pancreas (PatuT) and lung carcinoma (A549). All cell lines feature significant binding of MR-Fc and mAb100 to the cell surface compared to all controls (bar A,C-E and G, respectively)

amounts of MR-Fc ligands localized intracellular and/or on the plasma membrane, compared to controls. To identify these glycoprotein ligands, and to characterize their glycan components, MR-Fc was used as a probe for the enrichment of ligands from A549 whole cell lysates, followed by trypsin treatment and subsequent glycoproteomics using LC–MS. We also attempted to specifically identify cell surface MR-Fc ligands using a biotinylation strategy with intact cells; however, we were not successful in achieving quantitative and selective biotinylation of only cell surface glycoproteins. This approach, while perhaps appealing, is complicated by the lack of established technologies, and also by the dynamic nature of the movement of intracellular glycoproteins to and from the plasma membrane.

Glycoprotein ligands were isolated from total cell lysate in the presence of calcium, since mannose binding of the CTLD is calcium-dependent. In addition, the enrichment was performed in the presence of EDTA to identify non-specific binding. Intact glycopeptides were analyzed from this dataset to identify MR-Fc glycoprotein ligands. In total 330 N-glycopeptides were identified on 115 N-glycosylation sites in addition to 16 O-glycopeptides originating together from a total of 42 glycoproteins. Seven of them are ER-located glycoproteins and 13 lysosomal enzymes, next to 22 glycoproteins that predicted to be localized on the cell membrane, in the extracellular space or as secreted glycoproteins (Table 1, Suppl. Table S7). Of the latter group, nine glycoproteins are also potentially present in the lysosome based on Uniprot annotation.

Strikingly, the identified glycoproteins carried almost exclusively paucimannose ($Man_{1-3}GlcNAc_2Fuc_{0-1}$) or oligomannose ($Man_{4-9}GlcNAc_2$) N-glycans (Fig. 4, Table 1, Suppl. Table S7). Together they account for more than 90% of the glycoforms present on all identified N-glycosylation sites (Fig. 4b). While $Man_3GlcNAc_2Fuc_1$ is the main paucimannose N-glycan, $Man_6GlcNAc_2$ was the oligomannose N-glycan present on most N-glycosylation sites (Fig. 4b). In addition, core fucosylated $Man_4GlcNAc_2Fuc_1$ and $Man_5GlcNAc_2Fuc_1$ were found on lysosomal alpha-glucosidase and lysosome-associated membrane glycoprotein 1. Phospholipase D3 and putative phospholipase B-like 2 were the only proteins that lacked oligomannose N-glycans and instead carried paucimannose N-glycans. A few glycopeptides originating from phospholipase D3, lysosomal α-mannosidase, CD63 antigen, lysosomal acid phosphatase and galectin-3-binding protein, were identified as also expressing hybrid and complex N-glycans (Table 1). In summary, the N-glycan distribution analyzed overall identified glycosylation sites reveals that the majority of all MR-Fc glycoprotein ligands carry pauci- and oligomannose N-glycans (Fig. 4b), which is in accordance with the glycan-preference of the MR deduced from the microarray data (Fig. 3b,c).

Interestingly, we observed O-glycopeptides on the lysosomal enzymes beta-glucuronidase, lysosomal acid

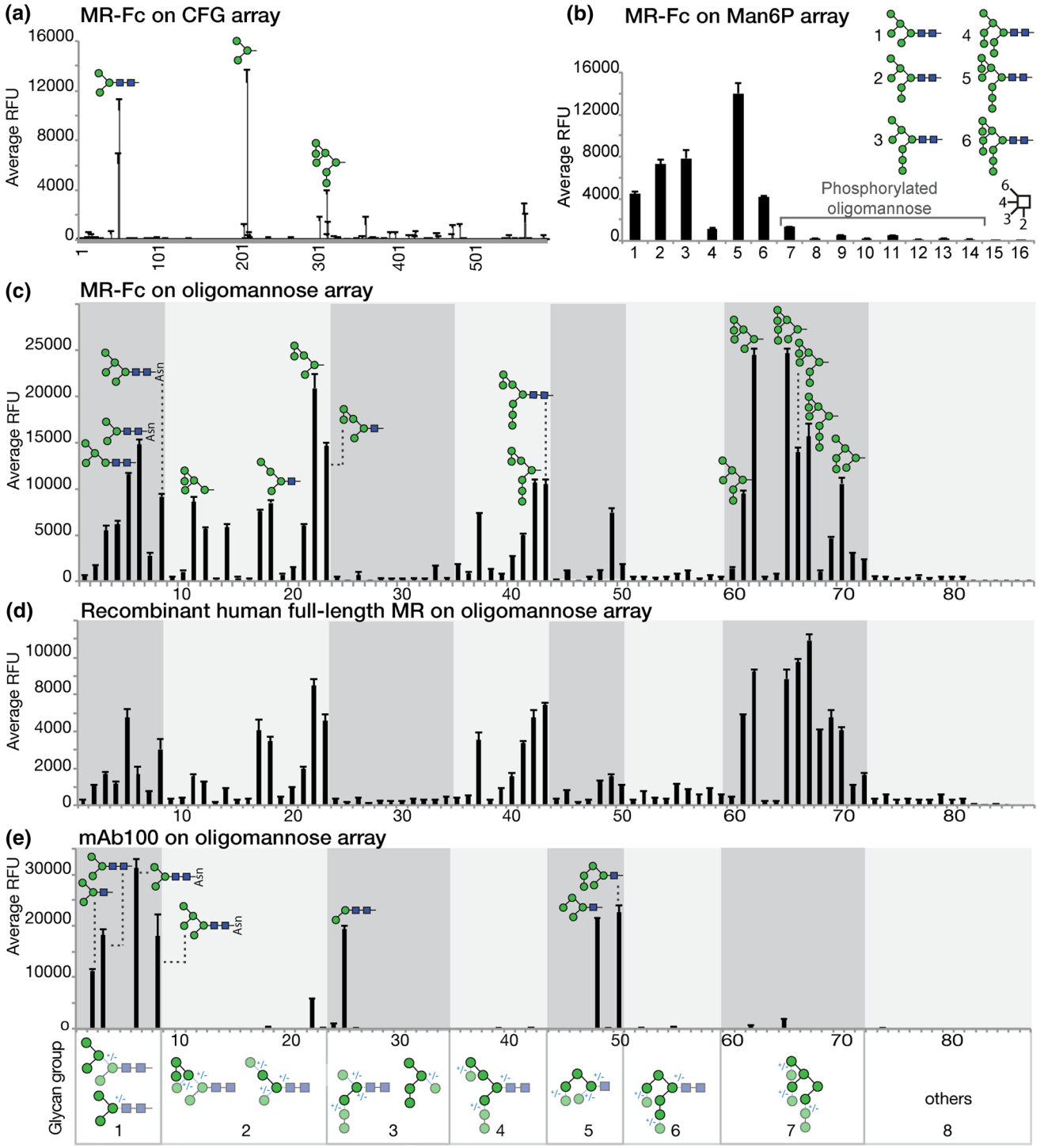


Fig. 3 Identification of MR-Fc glycan ligands using glycan microarrays. The major endogenous ligands of MR-Fc are mannose-containing N-glycans covering a broad range of pauci- and oligomannose N-glycans. **a** CFG microarray probed with MR-Fc. **b** Mannose-6-phosphate array probed with MR-Fc. **c** Oligomannose array probed with MR-Fc, **d** Oligomannose array probed with the full-length recombinant human MR, showing identical binding as MR-Fc. **e** Oligomannose array probed with mAb100. Glycan group 1: glycans with a Manα1-6(Manα1-3)Man motif. Glycan group 2: glycans with the extension of the 6-arm of the glycan core. Glycan group 3: glycans with the extension of the 3-arm of the glycan core. Glycan group 4: glycans with a Manα1-6Man or a Manα1-2Manα1-6Man extension of the 6-arm of the glycan core, while the 3-arm is extended with one or two Manα1-2Man. Glycan group 5: glycans with a Manα1-3Man or a Manα1-2Manα1-3Man extension of the 6-arm of the glycan core. Glycan group 6: glycans with a Manα1-3Man or a Manα1-2Manα1-3Man extension of the 6-arm of the glycan core, while the 3-arm is extended with one or two Manα1-2Man. Glycan group 7: different isoforms of Man6-Man9. Glycan group 8: other mannose-containing and complex glycans. Glycan linkages are indicated based on the Oxford system [70]

Table 1 MR-Fc glycoprotein ligands in A549 lung cancer cell lines

Uniprot Accession	Gene name	Protein name	Identified N-glycosylation sites	Paucimannose	Oligomannose	Oligomannose-Fucose	Hybrid / complex	O-glycan	Subcellular localization
O00469	PLOD2_HUMAN	Procollagen-lysine,2-oxoglutarate 5-dioxygenase 2	N209, N522, N696						Rough ER membrane
O95479	G6PE_HUMAN	GDH/6PGL endoplasmic bifunctional protein	N282						ER lumen
Q02809	PLOD1_HUMAN	Procollagen-lysine,2-oxoglutarate 5-dioxygenase 1	N538						Rough ER membrane
Q15155, Q5JPE7, P69849	NOMO1_HUMAN, NOMO2_HUMAN, NOMO3_HUMAN	Nodal modulator 1, Nodal modulator 2, Nodal modulator 3	N218, N618						ER membrane
Q96AY3	FKB10_HUMAN	Peptidyl-prolyl cis-trans isomerase FKBP10	N70, N182, N294, N310, N407						ER lumen
Q9Y4L1	HYOU1_HUMAN	Hypoxia up-regulated protein 1	N515, N596, N830, N862, N869						ER lumen
Q8IV08	PLD3_HUMAN	Phospholipase D3	N387						ER, Lysosome, Golgi apparatus, Endosome
O00754	MA2B1_HUMAN	Lysosomal alpha-mannosidase	N133, N367, N651, N692, N766						Lysosome
P06865	HEXA_HUMAN	Beta-hexosaminidase subunit alpha	N157						Lysosome
P08236	BGLR_HUMAN	Beta-glucuronidase	N173, N272						Lysosome
P10253	LYAG_HUMAN	Lysosomal alpha-glucosidase	N140, N390, N470, N882, N925						Lysosome
P11117	PPAL_HUMAN	Lysosomal acid phosphatase	N92, N133, N167, N177, N191, N267, N331						Lysosome
P15586	GNS_HUMAN	N-acetylglucosamine-6-sulfatase	N183, N210, N198, N317, N362, N387, N405						Lysosome
P22304	IDS_HUMAN	Iduronate 2-sulfatase	N115, N280						Lysosome
P54802	ANAG_HUMAN	Alpha-N-acetylglucosaminidase	N261						Lysosome
Q8NHP8	PLBL2_HUMAN	Putative phospholipase B-like 2	N515						Lysosome
Q9UBR2	CATZ_HUMAN	Cathepsin Z	N184						Lysosome
Q9UHG3	PCYOX_HUMAN	Prenylcysteine oxidase 1	N196, N353						Lysosome
Q9UMR5	PPT2_HUMAN	Lysosomal thioesterase PPT2	N206						Lysosome
O15230	LAMA5_HUMAN	Laminin subunit alpha-5	N2209, N2423, N2707						Basement membrane
O60568	PLOD3_HUMAN	Multifunctional procollagen lysine hydroxylase and glycosyltransferase LH3	N548						ER, secreted, extracellular space
O75882	ATRN_HUMAN	Attractin	N383, N428						Cell membrane, secreted
P02452	CO1A1_HUMAN	Collagen alpha-1(I) chain	N1365						Extracellular matrix
P02458	CO2A1_HUMAN	Collagen alpha-1(II) chain	N1388						Extracellular matrix
P02786	TFR1_HUMAN	Transferrin receptor protein 1	N727						Cell membrane, melanosomes
P06276	CHLE_HUMAN	Cholinesterase	N85, N284, N369						Secreted
P06731	CEAM5_HUMAN	Carcinoembryonic antigen-related cell adhesion molecule 5	N197, N204, N246, N375, N480, N553						Plasma membrane, cell surface
P07339	CATD_HUMAN	Cathepsin D	N263						Lysosome, melanosome, extracellular space
P07602	SAP_HUMAN	Prosaposin	N80, N215						Lysosome, secreted

phosphatase, N-acetylglucosamine-6-sulfatase, next to lysosome-associated membrane glycoprotein 1, the transferrin receptor protein 1 and mucin 5-B (Table 1, Suppl. Table S7). Our data strongly suggests that these glycopeptides were mainly modified with the Tn antigen (GalNAcα1-Ser/Thr), which was determined based on the diagnostic oxonium ion ratio for GalNAc-containing glycans using HCD fragmentation [41]. The transferrin receptor protein 1 was the

Table 1 (continued)

P08962	CD63_HUMAN	CD63 antigen	N130						Cell membrane, lysosome, melanosome, extracellular exosomes, endosome
P11047	LAMC1_HUMAN	Laminin subunit gamma-1	N650, N1107, N1205, N1223, N1241, N1395						Basement membrane
P11279	LAMP1_HUMAN	Lysosome-associated membrane glycoprotein 1	N62, N76, N84, N103, N261						Lysosome, endosome, plasma membrane
P13473	LAMP2_HUMAN	Lysosome-associated membrane glycoprotein 2	N101, N257, N356						Lysosome, endosome, plasma membrane
P40199	CEAM6_HUMAN	Carcinoembryonic antigen-related cell adhesion molecule 6	N152, N197, N224, N256						Plasma membrane, cell surface
P43251	BTD_HUMAN	Biotinidase	N119, N150, N203, N349						Extracellular space
P50897	PPT1_HUMAN	Palmitoyl-protein thioesterase 1	N197, N212, N232						Lysosome, secreted
Q08380	LG3BP_HUMAN	Galectin-3-binding protein	N69, N125, N398, N551						Secreted, extracellular matrix
Q13510	ASAH1_HUMAN	Acid ceramidase	N259						Lysosome, secreted
Q92542	NICA_HUMAN	Nicastrin	N45, N435						Membrane, melanosome
Q92820	GGH_HUMAN	Gamma-glutamyl hydrolase	N203						Lysosome, melanosome, extracellular space
Q9HC84	MUC5B_HUMAN	Mucin-5B	N145, N1774, N4804, N4960, N5046						Secreted
Q9UHL4	DPP2_HUMAN	Dipeptidyl peptidase 2	N315, N428						Lysosome, cytoplasmic vesicles, secreted

A549 glycoproteins were immunoprecipitated using MR-Fc in a calcium-dependent manner. Intact glycopeptides were analyzed using glycoproteomics. All identified N-glycosylation sites (column 4) are listed, with their corresponding glycan classes summed per protein (column 5–9). The subcellular protein localization is based on Uniprot

only protein with a glycan composition corresponding to disialylated core 1 O-glycan. Glycopeptides with the glycan portion $Hex_2HexNAc_2$ were all classified as N-glycans since the oxonium ion ratio indicates the presence of GlcNAc and the fragmentation pattern of glycopeptide B- and Y-ions is indicative for N-glycans. Most of these glycopeptides were also detected to express the glycan $Hex_2HexNAc_2Fuc_1$, with the fucose attached to the innermost GlcNAc, confirming N-glycan core fucosylation.

Discussion

Here we identify pauci- and oligomannose-type N-glycans as the major endogenous glycan ligands of the mannose-recognizing domains (CTLD4-7) of the MR, as present in MR-Fc. Ligands of MR-Fc displayed a cell- and tissue-specific distribution in a variety of tumor and control tissues. Interestingly, microarray analysis of the MR glycan-ligands, and glycoproteomics of the lung cancer cell line A549 revealed a novel set of pauci- and oligomannose-carrying glycoprotein ligands recognized by the MR. Our study contributes significantly to our knowledge of the glycan-specificity and preference of the MR. Moreover, it suggests a hitherto unexplored potential role of pauci- and oligomannose-carrying glycoproteins and the MR in tumor recognition and immune evasion.

CTLD4-7 of the MR have been reported to bind mannose and fucose, and with lower affinities to both GlcNAc and glucose [5–7]. Most of those binding studies were performed on immobilized monosaccharides or with yeast mannans. Using glycan microarray technology, we screened MR-Fc binding to more than 700 mammalian glycans, and revealed that pauci- and oligomannose N-glycans are the main endogenous MR-Fc glycan ligands. Our conclusion was corroborated by an orthogonal approach using glycoprotein immunoprecipitation and subsequent glycopeptide analysis. These data demonstrate that MR-Fc binds to many pauci- and oligomannose N-glycans, with compositions ranging from $Man_2GlcNAc_2$ to $Man_9GlcNAc_2$, including their core-fucosylated glycoforms. In addition, the binding patterns of the murine MR-Fc and full-length recombinant human MR were similar, making MR-Fc a suitable probe to investigate human MR ligands. Based on our results from the glycan microarray studies, and glycoprotein immunoprecipitation, we did not observe MR-Fc binding to any alternative endogenous glycans in human cells beyond these mannose-terminating structures. In particular, no MR-Fc binding was

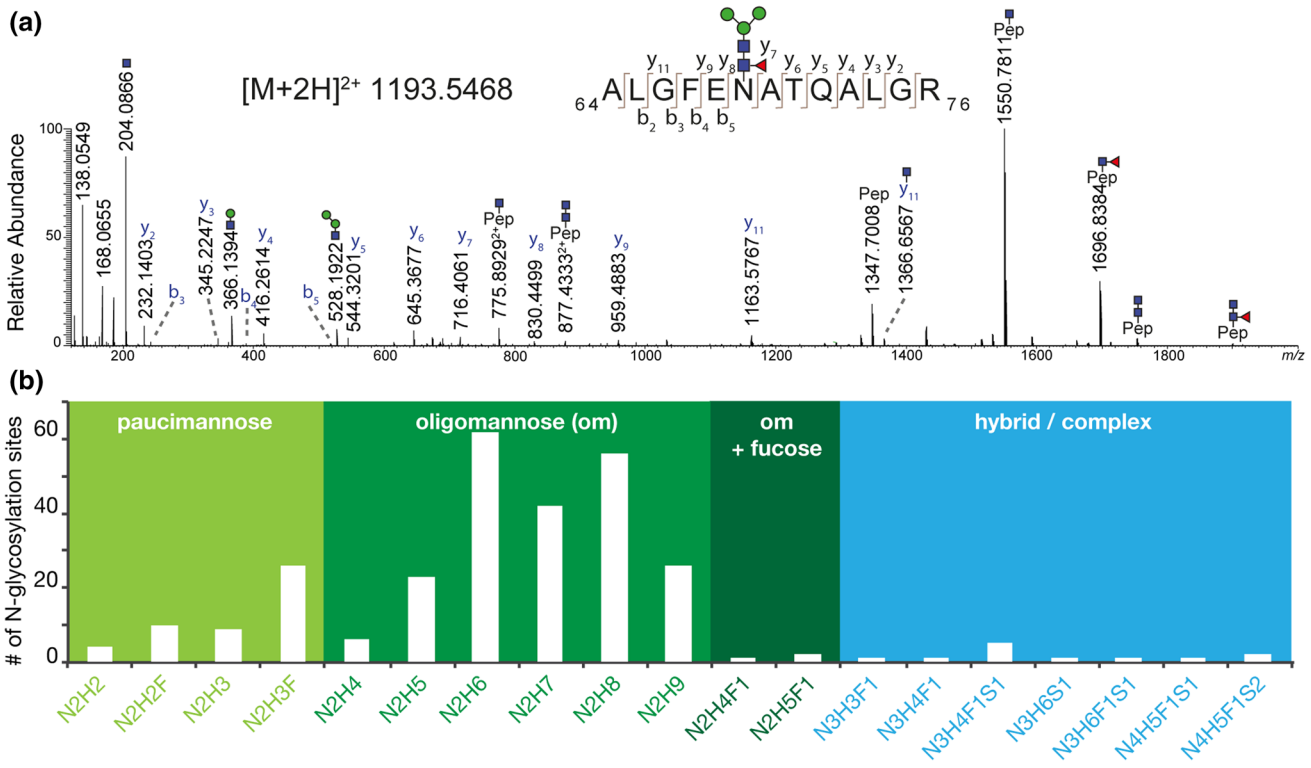


Fig. 4 Identification of glycoprotein ligands by glycoproteomics. **a** Representative HCD MS^2 spectra of a paucimannose-containing tryptic N-glycopeptide of galectin-3-binding protein (Q08380). **b** N-glycan distribution per number of glycosylation sites showing that the majority of all N-glycosylation sites carry pauci- and oligomannose glycans

observed to several fucosylated glycans, such as Lewis and blood group antigens, nor to terminal GlcNAc-containing glycans in different presentations on the CFG array. Overall these data indicate that fucose- and GlcNAc-containing endogenous glycans are not high-affinity ligands for the MR. Considering the binding of MR-Fc to $GlcNAc_3$-AEAB on the *S. mansoni* glycan array (Suppl. Table S5), it is possible that the MR may require multimeric or multivalent presentations of glycans with either GlcNAc or Fuc to enable lower affinity interactions, consistent with our observation that the MR-Fc can bind to high-density GlcNAc-agarose beads. Notably, such multivalent presentation of fucose or GlcNAc has been observed on pathogens, more so than on endogenous human glycan structures.

The initial screening of a tissue microarray by IHC revealed that the MR-Fc binds to distinct cancer and control tissues expressing pauci- and oligomannose-type N-glycans. MR-Fc ligands were abundantly detected in some control tissues, such as liver hepatocytes and kidney tubular endothelium. Using the paucimannose-specific antibody mAb100 we confirmed that a subset of these MR-Fc ligand-containing cell in most tissues contain paucimannose N-glycans next to oligomannose types. Our studies are consistent with the presence of MR-Fc ligands in several mouse tissues [27]. We observed very similar staining of immune cells in the human lung, which most likely originates from alveolar macrophages. Our recent identification of the human lung N-glycome revealed high levels of oligomannose-type compared to complex-type N-glycans [42]. It is well known that macrophages contain large amounts of lysosomal enzymes, which may include phosphorylated oligomannose-type N-glycans for trafficking to the lysosome via the mannose-6-phosphate receptor [43]. Upon removal of the phosphate residues after entering the lysosome [44], the underlying oligomannose N-glycans then present as intracellular ligands for the MR-Fc in IHC. MR-Fc binding to immune cells was also detected in a variety of other tissues, potentially accounting for tissue-resident macrophages. We further observed MR-Fc staining of epithelial cells in colon, prostate, bladder (urothelium) and kidney tubules. Human vaginal [45] and pigment [46] epithelium have been reported to express the MR, next to endothelial cells in several tissues [27]. Liver sinusoidal endothelial cells express the MR to endocytose glycoproteins, e.g. lysosomal enzymes, and remove them from the circulation [28]. This might explain an accumulation of intracellular mannose-containing glycoproteins inside these cells. Pancreas tissue showed moderate binding of MR-Fc to acini cells and to a lower extent to islet cells, which is in line with previous reports [27], and could be attributed to oligomannose-type N-glycans in RNase B

[7]. The wide distribution of MR-Fc ligands in control tissue is in agreement with the involvement of MR in homeostasis, by targeted uptake of lysosomal enzymes by different types of cells, e.g., endothelial cells [28] or clearance of proteins from the circulation or at sites of inflammation by immune cells [47]. However, the tissue microarray data also suggest the presence of cell surface-bound MR-Fc ligands. Further investigation of the glycoprotein ligands in these tissues will give new insights to enhance understanding of the role of the MR in homeostasis.

Recently, relatively high-level expression of pauci- and oligomannose-type N-glycosylation signatures have been detected in a variety of cancer tissues and cancer cell lines, suggesting their involvement in cancer biology [11, 12]. In our study, the three lung cancer and three control tissues display the most differential binding among all tissue types. While MR-Fc ligands were only moderately expressed in control lung tissue, our data showed strong binding of MR-Fc to the cell surface of lung cancer cells.

As we observed that the MR could bind to surface glycoproteins of A549 cells, we used immunoprecipitation and glycoproteomics to identify glycoproteins interacting with the MR-Fc. In the lung cancer cell line A549 we identified 42 glycoproteins, all carrying at least one N-glycosylation site with either pauci- or oligomannose glycans attached. A recent study identified more than 75% pauci- and oligomannose-containing N-glycans in A549 cells, which might contribute to the intense binding of MR-Fc we observed to the A549 cell surface (**Fig. 2**) [11].

On a glycopeptide and glycoprotein level we compared our data set with a glycoproteomics study of the non-small cell lung cancer (NSCLC) cell line PC9-IR [48]. Using ConA-lectin enrichment the latter study identified 60% of the pauci- and oligomannose-containing glycoproteins that we found to be MR-Fc glycoprotein ligands. Compared to a data set from breast cancer xenograft tissue approximately 83% of the MR-Fc ligands in A549 were found to express pauci- or oligomannose N-glycans [49]. Active paucimannose-containing glycoproteins, e.g. beta-glucuronidase, LAMP2 and CEACAM6, were also shown to be secreted by neutrophils upon exposure to pathogens such as *Pseudomonas aeruginosa* [50]. Thus, the results of our glycopeptide characterization, focused on MR-ligands, are largely in agreement with those of more general cancer glycomics and glycoproteomics investigations. In addition, this further supports our finding that the MR can bind ligands on different cancer cell types.

Interestingly, lung control tissue shows the highest levels of MR mRNA among 55 different tissue types, which was mainly attributed to the presence of alveolar macrophages (www.proteinatlas.org) [51, 52]. These macrophages account for approximately 90% of the leukocytes in a healthy lung where they play an important role in alveolar homeostasis [53]. This might raise the question whether tumor onset and progression are promoted by tumor cells presenting MR ligands in a tissue environment rich in MR-positive macrophages.

During tumorigenesis circulating monocytes and tissue-resident macrophages infiltrate into the tumor microenvironment and can differentiate into TAMs. In early tumor stages, these TAMs are mainly pro-inflammatory M1-like macrophages, which play an important role in immune recognition and clearance of tumor cells and have been associated with a positive outcome. At the later stage of tumor progression this polarization shifts towards more anti-inflammatory M2-like macrophages that promote growth, angiogenesis and immune evasion [18, 54]. Consequently, TAMs have been correlated with poor patient prognosis in cancer [55], including lung cancer [20]. The MR is an established marker of these alternatively activated M2-polarized macrophages [21, 22]. $CD14^{+}$-MR^{+} macrophages have also been identified as a diagnostic marker to differentiate benign pleural effusions from malignancy, including adenocarcinoma, squamous cell carcinoma and small cell lung cancer [56].

The MR has been reported to regulate suppressive immune responses [10, 14]. Endogenous ascitic fluid components isolated from ovarian cancer can inhibit the internalization of particles in TAMs via the MR [57], and can induce an increase in IL-10 secretion and a reduction in CCL3, suggesting a role of the MR in inducing an anti-inflammatory phenotype. Other lectin receptors, such as DC-SIGN, MGL and SIGLECs, have similarly been shown to promote tumor progression by inducing anti-inflammatory cytokine release upon receptor-glycan interaction (recently reviewed in [13]). Thus, it can be hypothesized that the pauci- and oligomannose-containing cell surface glycoproteins expressed on tumor cells bind to the MR on TAMs, thereby inducing an anti-inflammatory environment, which promotes tumor progression.

The involvement of MR on macrophages or dendritic cells in tumor development has been proposed in follicular lymphoma (FL). Surface Ig in FL are characterized by a high amount of Fab glycosylation in the variable domain, predominantly carrying oligomannose-type N-glycans [58]. Binding of the MR-Fc to FL cells, but not in normal B cells, can induce signaling, which can be blocked by preincubation with the mannosylated single-chain variable fragment (scFv). The authors speculate that MR-mediated B cell receptor signaling provide FL B cells alternative survival cues, independent of antigen-binding signals from the germinal center [59].

Interestingly, among the MR-Fc ligands in A549 lung cancer cell lines, known to be secreted or expressed on the cell surface, are GPI-anchored cell adhesion molecules CEACAM5 and CEACAM6. They are abundantly expressed in lung cancer tissues [60], on the cell surface of A549 cell

lines [61] and function as markers for different cancers [62]. CAECAM6 was identified as a regulator of A549 cancer cell proliferation [63]. Moreover, CEACAMs can activate Kupffer cells in liver metastasis, inducing the secretion of anti-inflammatory cytokines that promote tumor survival [62, 64]. The secretion of heavily glycosylated Muc5B is an important feature in lung homeostasis, maintaining mucociliary clearance of invading pathogens [53]. While we identified Muc5B as a ligand of MR-Fc, mucin gene mRNA was found to be overexpressed in lung cancer and specifically Muc5B and Muc5Ac mRNA expression levels were correlated with tumor recurrence [65]. In addition, a large portion of the MR glycoprotein ligands in A549 lung cancer cell line are lysosomal enzymes, e.g. cathepsin Z and cathepsin D, which could be secreted by the cell. TAMs have been reported to secrete proteolytically active cathepsin B and S to remodel extracellular matrix components and thus, promoting tumor growth [66]. Similarly, cathepsin Z has been found in cancer cells, including A549, suggesting its involvement in malignancy [67]. Furthermore, the MR-Fc ligands, lysosome-associated membrane proteins (LAMPs), play an important role in tumor biology [68]. Thus, many of the MR-Fc ligands identified in the A549 cells are associated with the progression of lung cancer, which may indicate the involvement of the MR, and opens up novel possibilities to study the functional role of the MR in lung cancer.

In summary, we present the first detailed study on the glycan structural features of the endogenous ligands for MR CTLD4-7 and have mapped these ligands across a wide range of tissues, including cancer and control tissue. While the tissue- and cell-specific presence of MR-Fc ligands in control tissue is linked to its involvement in homeostasis, MR-Fc binding to cancer tissue and cell lines suggests a functional role of the MR in cancer biology. Further studies of the specific glycoprotein ligands in these tissues will reveal novel insights into the role of MR in homeostasis and cancer biology. Of particular interest will be the investigation of a potential MR-dependent anti-inflammatory tumor microenvironment induced by mannose-presenting tumor cells, e.g. in lung adenocarcinoma. Detailed knowledge about MR glycan ligands on the other hand might also facilitate research to therapeutically target and deplete M2 polarized macrophages in the tumor microenvironment.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00018-021-03863-1.

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Author contribution KS, LCL and CG performed the experiments. AYM and JHM supported the microarray experiments and analysis. JNG assessed the tissue pathology. KS, IVD, RDC conceptionally designed the work and wrote the manuscript. All authors read and approved the final manuscript.

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Availability of data The MS proteomics data have been deposited to the ProteomeXchange Consortium (http://proteomecentral.proteomexchange.org) via the PRIDE partner repository [69] with the dataset identifier PXD022546. Glycan microarray data are available in supplementary material and at the NCFG website (https://ncfg.hms.harvard.edu/ncfg-data).

Declarations

Conflict of interest The authors declare no conflicts of interest.

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